

Strings Methods

⚠ 1. Strings Are Immutable

You **cannot change** a string directly.

```
let name = "Prajwal";
```

```
name[0] = "K";
```

```
console.log(name); // Prajwal (unchanged)
```

Every method returns a **NEW STRING**.

🔥 This matters for performance + debugging.

1 slice(start, end)

Most used. Supports negative indexing.

```
let text = "JavaScript";
```

```
text.slice(0,4); // "Java"
```

```
text.slice(-6); // "Script"
```

✓ Does NOT modify original

✓ Accepts negative values

Use Case: Substrings in APIs, parsing.

2 substring(start, end)

Similar to slice BUT ❌ no negative support.

```
text.substring(0,4); // "Java"
```

If negative → treated as 0.

4 indexOf()

```
let msg = "I love coding";
```

```
msg.indexOf("love"); // 2
```

```
msg.indexOf("z"); // -1
```

5 lastIndexOf()

```
"hello hello".lastIndexOf("hello"); // 6
```

6 includes() Modern Way

```
msg.includes("love"); // true
```

7 startsWith() / endsWith()

```
msg.startsWith("I"); // true
```

```
msg.endsWith("ing"); // true
```

8 toUpperCase() / toLowerCase()

```
"user".toUpperCase(); // USER
```

9 trim() / trimStart() / trimEnd()

Removes whitespace.

```
" hello ".trim(); // "hello"
```

10 replace()

```
let s = "I love JS";
```

```
s.replace("JS","Node"); // I love Node
```

1 **1** replaceAll()

```
"ha ha ha".replaceAll("ha","ho"); // ho ho ho
```

1 **2** split()

Converts string → array.

```
"apple,banana,mango".split(",");
```

Output:

```
["apple","banana","mango"]
```

1 **3** concat()

```
"Hello".concat(" ","World");
```

```
"Hello" + " World"
```

1 **4** charAt(index)

```
"Hello".charAt(1); // e
```

Same as:

```
"Hello"[1]
```

1 **5** charCodeAt()

Returns ASCII / Unicode.

```
"A".charCodeAt(0); // 65
```